RESUME

Industrial & Systems Engineer (Ph.D.) | Simulation Solution Consultant | AI Solution Architect

JEHUN LEE

Address: Yongin, Gyeonggi-do, South Korea
Mobile: (+82) 10-8244-5376 **Portfolio:** https://jehun-lee.work
Email: jehun.lee302@gmail.com, jehun.lee@kaist.ac.kr
Linkedin: https://www.linkedin.com/in/jehun-lee/
Scholar: https://scholar.google.com/citations?user=C7ekyjEAAAAJ

Professional Profile

In the AI era, the most important question is not which model to use, but what problem is worth solving — defined through human will and necessity. Algorithms don't choose direction; people do.

My direction is to be a keyman: someone who sees the entire system, maps stakeholders, sets priorities, and deeply understands the problems that need solving so direction can be defined. The growth that lasts is the growth where the organization grows together — not where individuals optimize alone.

Systemic excellence, to me, means hiring AI agents — treating them as employees who execute autonomously, with humans setting direction and confirming key decisions. Chores, basic reviews, missing numbers move into agent pipelines, freeing human attention for the work only humans can do: judgment, strategy, meaning.

Education

2021.03 - 2025.02	**Ph.D. in Industrial & Systems Engineering**	Korea Advanced Institute of Science and Technology (KAIST)
2016.09 - 2021.02	**M.S. in Industrial Engineering**	Sungkyunkwan University (SKKU)
2013.03 - 2016.08	**B.S. in Systems Management Engineering**	Sungkyunkwan University (SKKU)
2011.03 - 2013.02	**High School in Math & Science Track**	Sejong Science High School

Work Experience

2025.08 - Present

Planning & Scheduling Solution Consultant
VMS Solutions Inc. | Solution Division | Consultant Part
Ministry of National Defense | Tech. Research Personnel [2025.08 - 2026.02]
System designer & PMO: digital twin platform, production-logistics integrated simulator, simulation-based AI scheduler, SaaS product
– SK Hynix Digital Twin Platform: production-logistics simulator PM, use-case discovery, KPI definition, data consistency metrics, MES legacy data analysis & simulator integration, AI prediction model data specification
– Production simulation engine architecture redesign & development participation
– Planned and authored 2 government R&D proposals on Agentic AI (manufacturing-specific Agentic AI, LLM-based production plan explanation module / with University of Toronto)
– AI extensibility review & design for simulation-based APS, new engine feature definition, participation in GenAI-based development
– LLM-based schedule analysis & report generation system performance review and feedback
– 2 SaaS solution consulting engagements; technical review supporting client executives' strategic decisions (general manufacturing, semiconductor back-end)

2025.02 - 2025.07

Planning & Scheduling Solution Consultant
VMS Solutions Inc. | Solution Division | Micron Part
Ministry of National Defense | Tech. Research Personnel [2025.02 - 2025.07]
Solution consultant: simulation-based real-time APS maintenance, system configuration remote support
– Micron global fab simulation-based real-time APS (advanced production and scheduling) system logic maintenance and remote system configuration support
– Production-logistics integrated simulation design

2021.03 - 2025.02 **Researcher**
KAIST | MSS (Manufacturing Service Systems) Lab.
Ministry of National Defense | Tech. Research Personnel [2023.03 - 2025.02]
Lead researcher: AI-based dynamic schedulers, optimization-based heuristics, autonomous manufacturing scheduling systems
– 9 industry-academia projects (planned 5, PM 4), 4 government projects (planned 3, PM 4)
– 2 SCI journal papers (1 first-author, 1 co-author), 11 international conference presentations (8 first-author, 3 co-author)
– Fundamental research on Neural Combinatorial Optimization based on GNN and RL
– Designed general-purpose scheduling agent and Meta-RL methodology for flexible adaptation to manufacturing environment changes
– Designed AI-based demand forecasting, production planning, and job scheduling optimization algorithms; validated business impact using real production data (Samsung Electronics, LG Electronics, Samsung Heavy Industries, etc.)
– Provided technical review supporting client executives' strategic decisions and delivered system design proposals

2016.04 - 2021.02 **Researcher**
SKKU | SCO (Systems Control & Optimization) Lab.
Scheduling algorithm developer: meta-heuristics, optimization; System designer: digital twin for manufacturing, dynamic rescheduling systems
– 5 industry-academia projects (planned 2, PM 2), 4 government projects (planned 2, PM 1)
– 3 SCI journal papers (1 first-author, 2 co-author), 10 international conference presentations (5 first-author, 5 co-author)
– Led foundational design and operations optimization research for 3D printer-based smart factory digital twin systems
– Developed rescheduling algorithms for handling disruptions (machine failures, order cancellations) in flexible manufacturing systems (FMS)
– Translated physical manufacturing constraints into mathematical models and validated in virtual environments
– Defined virtual factory operation scenarios and conducted simulation-based operational efficiency reviews

Projects

2026.02 - Present (4m) **Digital twin platform construction: production-logistics simulation-based operation twin for semiconductor fab**
VMS Solutions Inc. (with SK Hynix, SKT, SK AX, Calro)

2025.06 - 2025.12 (7m) **Digital twin platform construction: production-logistics simulation-based operation twin for semiconductor fab (PoC)**
VMS Solutions Inc. (with SK Hynix, SKT, SK AX, Calro)

2025.03 - Present (16m) **Software product design/development: simulation-based scheduling for semiconductor manufacturing**
VMS Solutions Inc.

2025.02 - 2025.06 (5m (5y)) **Simulation-based real-time scheduler for semiconductor fab**
VMS Solutions Inc. (with Micron)

2024.05 - 2025.02 (10m (10y)) **Autonomous scheduling for swift and efficient adaptation to dynamic manufacturing environments**
KAIST (with NRF of Korea)

2024.03 - 2025.02 (1y) **AI-based algorithm for optimal operation plans for various scenarios**
KAIST (with Samsung Electronics)

2023.07 - 2024.06 (1y) **Reinforcement learning for unrelated parallel machine scheduling problems with sequence-dependent setup times and machine eligibility**
KAIST (with VMS Solutions)

2023.07 - 2024.02 (8m) **Production planning with AI**
KAIST (with LG Electronics)

2022.05 - 2023.02 (10m) **Graph-based reinforcement learning algorithm for real-time job shop scheduling**
KAIST (with NRF of Korea)

2022.04 - 2023.03 (1y) **Reinforcement learning for job shop scheduling**
KAIST (with VMS Solutions)

2022.03 - 2022.03 (1m (7m)) **Development of a wiring optimization algorithm for X-DEC slim layout**
KAIST (with SK Hynix)

2022.03 - 2025.02 (3y) **Reinforcement learning-based meta-scheduling for manufacturing systems**
KAIST (with NRF of Korea)

2022.03 - 2023.07 1y5m (2y) — **Reinforcement learning for project scheduling**
KAIST (with Samsung Heavy Industries)

2021.07 - 2021.11 5m — **Development of a reinforcement learning algorithm for workload balancing of ship cargo production**
KAIST (with Samsung Heavy Industries)

2020.06 - 2021.02 9m — **Optimal machine assignment with machine learning algorithms**
SKKU (with VMS Solutions)

2019.06 - 2022.05 3y — **Cyber-physical assembly and logistics systems in global supply chains**
KAIST (with MOTIE of Korea, Yura)

2019.09 - 2020.02 6m — **Optimal weight sets for dispatching rules with multiple KPIs**
SKKU (with VMS Solutions)

2019.04 - 2019.12 9m (1y9m) — **Big data-based simulation and optimization technology for smart manufacturing**
SKKU (with MOTIE of Korea, Samsung SDI)

2019.03 - 2022.02 3y — **Development of scheduling theory and algorithms with reinforcement learning for manufacturing systems**
SKKU (with NRF of Korea)

2018.07 - 2018.09 3m — **Methodology for dispatching rules’ weights**
SKKU (with SK Hynix)

2018.07 - 2019.02 8m — **Framework development for KPI analysis with various weights on dispatching rules**
SKKU (with VMS Solutions)

2017.07 - 2018.02 8m — **Analysis of KPIs according to weights on dispatching rules for LCD manufacturing**
SKKU (with VMS Solutions)

2017.07 - 2018.01 7m — **Design and analysis for operations optimizations of smart factory testbed**
SKKU (with MSIP of Korea)

2016.11 - 2019.10 3y — **Development of scheduling and rescheduling algorithms for 3D printer-based smart factory**
SKKU (with NRF of Korea)

2016.07 - 2017.02 8m — **Development of algorithms for detecting and improving inefficient schedules in LCD processes**
SKKU (with VMS Solutions)

2016.04 - 2018.05 2y2m (3y) — **Development of open FaaS IoT service platform for mass personalization**
SKKU (with MSIP of Korea)

Academic Works

2024 — **Graph-based imitation learning for real-time job shop dispatcher** — 1st Author
IEEE Transactions on Automation Science and Engineering [LINK]

2024 — **Tree-based Dispatcher for Job Shop Scheduling** — 1st Author
IEEE International Conference on Automation Science and Engineering (CASE)

2024 — **Tree-based dispatcher for solving job shop scheduling problems** — 1st Author
the Spring Conference of Korean Institute of Industrial Engineers

2023 — **Active schedule-based imitation learning for job shop scheduling** — 1st Author
the Spring Conference of Korean Institute of Industrial Engineers

2022 — **Imitation learning for real-time job shop scheduling using graph-based representation** — 1st Author
Winter Simulation Conference [LINK]

2022 — **Job shop scheduling using graph-based imitation learning** — 1st Author
INFORMS Annual Meeting

2022 **A multi-manned assembly line worker assignment and balancing problem with positional constraints** Co-Author [LINK]
IEEE Robotics and Automation Letters

2022 **Reinforcement learning for resource leveling in multiple projects** 1st Author
the Spring Conference of Korean Institute of Industrial Engineers

2022 **Dynamic job shop scheduling using graph-based imitation learning** 1st Author
the Spring Conference of Korean Institute of Industrial Engineers

2021 **Machine learning-based periodic setup changes for semiconductor manufacturing machines** 1st Author [LINK]
Winter Simulation Conference

2021 **Resource leveling in shipyard cargo hold process through reinforcement learning** Co-Author
the Autumn Conference of Korean Institute of Industrial Engineers

2021 **Assembly line worker assignment and balancing problem with positional constraints** Co-Author [LINK]
Advances in Production Management Systems (APMS)

2021 **Operation and optimization of the automotive parts assembly line considering worker skill levels** Co-Author
the Summer Conference of Korea CDE

2020 **A sequential search method of dispatching rules for scheduling of LCD manufacturing systems** 1st Author [LINK]
IEEE Transactions on Semiconductor Manufacturing

2020 **A simulation-based sequential search method for multi-objective scheduling problems of manufacturing systems** 1st Author
Winter Simulation Conference

2020 **Workforce assignment for automotive parts assembly lines** Co-Author
the Winter Conference of Korea CDE

2020 **Digital twin-based cyber physical production system architectural framework for personalized production** Co-Author [LINK]
The International Journal of Advanced Manufacturing Technology

2020 **Workforce assignment with a different skill level for automotive parts assembly lines** Co-Author [LINK]
Advances in Production Management Systems (APMS)

2019 **A sequential search framework for selecting weights of dispatching rules in manufacturing systems** 1st Author [LINK]
Winter Simulation Conference

2019 **A genetic algorithm for hybrid flow shop scheduling with multiple assembly stages** 1st Author
the Autumn Conference of Korean Institute of Industrial Engineers

2018 **Vulnerability analysis of evacuation transportation networks** Co-Author
International Journal of Industrial Engineering-Theory Applications and Practice

2018 **A framework for performance analysis of dispatching rules in manufacturing systems** 1st Author
Winter Simulation Conference

2018 **Rescheduling algorithms for 3D printer-based manufacturing systems** 1st Author
the Summer Conference of Korea CDE

2018 **Scheduling algorithms for 3D printer-based manufacturing systems** Co-Author
the Spring Conference of Korean Institute of Industrial Engineers

2017 **Rescheduling of flexible flow shop with sequence-dependent setup times and job splitting** Co-Author [LINK]
Winter Simulation Conference

2017 **3D printer based assembly process scheduling algorithm development** Co-Author
the Winter Conference of Korea CDE

Patents

2021.05.31 **Optimizing method for the manufacturing process** South Korea
A real-time optimization method for worker allocation in semi-automated assembly processes KR1020210070359
Applicants: KAIST, Yura (entire 12 people)

Honors & Awards

2025.11	**Second Prize, Ph.D. Thesis Competition** $200 Learning Schedulers for Job Shop Scheduling Problems	Korean Institute of Industrial Engineers (KIIE)
2023.12	**First Prize, 2023 Simulation Challenge** $3,000 Q-time-based Gating Control in Semiconductor Fabrication	Winter Simulation Conference (WSC), Micron
2023.09	**Third Prize, 2023 AI Competition: Solving Real-world Problems** $1,500 Integrated Planning Module using AI	Hankook & Company
2022.06 - 2023.02	**Ph.D. Candidate Research Incentive Support** $15,000	NRF of Korea
2022.09	**Second Prize, 2022 Poster Competition: Industry/Social Problems** RL for Resource Leveling in Shipbuilding	KAIST
2022.05	**Certificate of Appreciation: Successful Project** Workload Balancing of Ship Cargo Production	Samsung Heavy Industries
2021.03 - 2025.02	**Full-tuition Entrance Scholarship** Full Tuition Merit-based Entrance Scholarship (Ph.D.)	KAIST
2019.03, 2019.09	**Graduate Assistant Tuition Waiver** Half Tuition Tuition Waiver (M.S. & Ph.D Integrated Program)	SKKU
2018.03, 2019.03	**Simsan Scholarship** Half Tuition Scholarship (M.S. & Ph.D Integrated Program)	SKKU
2018.03	**Systems & Management Engineering Scholarship** Half Tuition Scholarship (M.S. & Ph.D Integrated Program)	SKKU
2017.03	**Industrial Engineering Alumni Association Scholarship** Half Tuition Scholarship (M.S. & Ph.D Integrated Program)	SKKU
2016.07	**Third Prize, 2015-2016 PACE RSMS Competition (Second Year): Customer Insight**	General Motors (GM)
2016.07	**Third Prize, 2015-2016 PACE RSMS Competition (Second Year): Manufacturing Engineering**	General Motors (GM)
2013.03 - 2016.08	**Full-tuition Entrance Scholarship** Full Tuition Merit-based Entrance Scholarship (B.S.)	SKKU

Activities & Leadership

2022.03 - 2023.08	**Researcher Representative**	Manufacturing and Service Systems Lab. (KAIST)
2021.10 – 2021.12	**Technical Mentor**	Public Data Internship Program (Korea National Information Society Agency)
2021.03 – 2024.12	**Teaching Assistant (TA)**	Scheduling (KAIST)
2018.09 – 2019.06	**Teaching Assistant (TA)**	Supply Chain Management (SKKU)
2018.03 – 2019.02	**President**	Industrial Engineering Graduate Student Council (SKKU)
2018.03 – 2018.06	**Teaching Assistant (TA)**	Operations Management (SKKU)
2016.10 – 2018.12	**Teaching Assistant (TA)**	Gender Awareness Education (SKKU)
2017.09 – 2017.12	**Teaching Assistant (TA)**	Engineering Economy (SKKU)
2017.03 – 2017.06	**Teaching Assistant (TA)**	Operations Research and Practice 1 (SKKU)
2016.09 – 2016.12	**Teaching Assistant (TA)**	Engineering Economy (SKKU)
2015.03 – 2016.08	**Working Group Member**	Systems Management Engineering Student Council (SKKU)
2015.03 – 2016.08	**President**	Turbo: Basketball Club (SKKU)
2014.08 – 2020.02	**Working Group Member**	Turbo: Basketball Club (SKKU)
2013.03 – 2015.02	**Working Group Member**	College of Engineering Student Council (SKKU)
2012.03 - 2013.02	**Head of Student Autonomy Association**	Student Council (High School, 11th Grade)
2010.03 - 2011.02	**Vice President of Student Council**	Student Council (Middle School, 9th Grade)

Skills

Programming	Python, C#, SQL, LaTeX, R
Methodologies	Reinforcement Learning (RL), Imitation Learning (IL), Behavior Cloning (BC), Supervised Learning, Unsupervised Learning, Meta-heuristics, Graph Neural Networks (GNN), Combinatorial Optimization
Frameworks & Libraries	PyTorch, Matplotlib, Pandas, NumPy
Optimization Solvers	Gurobi, CPLEX, OR-Tools
Simulation & Tools	VMS Mozart (APS), Plant Sim, MS Office, Photoshop
DevOps & Productivity	Git / GitHub, Docker, Claude Code, n8n, Notion
Languages	Korean (Native), English (Professional)